

Encrypted traffic classification based on contrastive learning with spatial-temporal feature fusion

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ABSTRACT

While encryption ensures the confidentiality and integrity of user data, more and more attackers try to hide attack behaviors through encryption, which brings new challenges to malicious traffic identification. How to effectively classify encrypted traffic without decrypting traffic and protecting user privacy has become an urgent problem to be solved. This paper proposes a classification method for encrypted traffic based on contrastive learning with spatial and temporal feature fusion. Unlike traditional contrastive learning with a single encoder, this method separately designs a spatial feature encoder and a temporal feature encoder to extract distinct features and implements contrastive learning in the same encoding layer through a fully connected layer to obtain a unique feature representation of encrypted traffic. This method focuses on the consistent features from different perspectives within the same sample. The effectiveness of this method is demonstrated through ablation experiments and comparisons with existing works.

Keywords: Encrypted traffic classification, spatial features, temporal features, contrastive learning

1. INTRODUCTION

Encrypted traffic, while fully protecting various aspects of internet users' security, also conceals various internet activities, which poses new challenges for traffic identification. With the progress of research on identifying encrypted traffic, supervised learning has been able to achieve satisfactory accuracy for specific tasks. However, in practical applications, the vast majority of traffic is unlabeled data, which limits the application scope of supervised methods. How to use large amounts of unlabeled data to train powerful and representative feature extraction models, is particularly suitable for encrypted traffic that loses semantic information and is difficult to manually annotate.

The various features of network traffic have consistency in high-dimensional representation space, which means that both spatial and temporal features of any traffic sample have consistency in the representation space. In this paper, based on the idea of self-supervised contrastive learning and inspired by image and text consistency representation tasks, we design a task of fusing the spatial and temporal features of encrypted traffic for encrypted traffic feature extraction. We propose a spatial and temporal dual encoder structure unlike previous ones to obtain representations of different features of encrypted traffic. The training objective is to obtain spatial and temporal representation consistency in a shared embedding space, and we use contrastive loss to fuse these two features. We transform the traffic classification task into a task of fusing spatial and temporal features and obtain consistent representations as the final traffic representation through mutual guidance between spatial and temporal features.

2. RELATED WORKS

2.1 Payload-based encrypted traffic classification

This method obtains information by analyzing the payload, establishes an encrypted traffic knowledge base, and matches specific fields in encrypted traffic data to achieve encrypted traffic discrimination. Özdel¹ proposed a payload-based application layer traffic identification method that can identify various application traffic. Tang² proposed a method that first assigns similar packets to the same group, and then uses a machine learning algorithm to classify each group. Bonfiglio et al.³ used Pearson's chi-square test and Naive Bayes classifiers to extract protocol frame structure fingerprints from the

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first 4 bytes of non-encrypted packet headers and payloads for Skype traffic identification. Shen⁴ used second-order Markov chain models to model traffic and considered application attributes such as port numbers and protocol types.

2.2 Statistical features based encrypted traffic classification

Statistical features focus on the characteristics of all packets within a certain period of time, such as the number of unidirectional packets, the time difference between the start and end of packet transmission, the frequency of packet transmission, etc. Then, the features are fed into a machine learning model for training to achieve the task of encrypted traffic identification. Mahdavi⁵ utilizes statistical features such as empirical distribution functions, mutual information, and autocorrelation functions to establish classifiers. Zhang⁶ uses features such as mutual correlation coefficients and wavelet analysis to establish classifiers. Wang⁷ uses statistical features such as wavelet analysis and mutual information to extract feature vectors, and uses a random forest classifier for classification. In addition, more than 20 flow statistical features such as session duration, inter-arrival time, packet size, and protocol fingerprints are often used in combination with Naive Bayes, support vector machines, decision trees, etc. for encrypted traffic identification⁸.

2.3 Machine learning based encrypted traffic classification

Researchers often combine machine learning with various feature to complete the task of traffic classification. According to different levels, encrypted traffic features can be divided into packet-level features, flow-level features, and session-level features. Bar-Yanai et al.⁹ proposed a new network traffic classification method that combines K-means and K-nearest neighbor algorithms. Moore et al.¹⁰ designed a network traffic feature classifier using supervised Bayesian estimation. Sun et al.¹¹ proposed a hybrid method for classifying encrypted traffic. This method first used signature matching to identify encrypted protocols such as IPsec and SSL/TLS, and then applied the Naive Bayes algorithm to identify encrypted applications. Alshammari¹² studied the performance of C5.0, AdaBoost, and genetic programming (GP) algorithms for identifying IP voice (VoIP) traffic. Besides payload features, temporal features have also been used as input for machine learning models. Gil et al.¹³ used temporal -related features such as flow duration, forward inter-arrival time, number of packets per second, and backward inter-arrival time, and employed k-nearest neighbor and C4.5 to classify network traffic. Yao et al.¹⁴ analyzed the discrete distribution and temporal patterns of flow features and proposed a hidden Markov model based on Gaussian Mixture.

2.4 Deep learning based encrypted traffic classification

Deep learning can automatically learn advanced features of data through training without manual design. Wang et al.¹⁵ proposed a classification framework based on CNN, comparing the experimental results between 1D-CNN, and 2D-CNN. Lotfollahi et al.¹⁶ proposed the Deep Packet traffic classification framework, and exceeded the SOTA results. Zou¹⁷ considered that there are effective features within continuous packets and used the characteristics of recurrent neural networks for learning. Liu et al.¹⁸ decomposed the traffic into many small-time windows and took the data flow of these time windows as the input of the network. The network used CNN and RNN to learn traffic sequence features and used fully connected layers for classification. Yao et al.¹⁹ first converted packets into feature vectors and input them into LSTM networks for encoding. This method shows that the combination of attention mechanisms and LSTM networks can effectively identify encrypted traffic. Shapira²⁰ converted traffic into a two-dimensional FlowPic image, and used CNN to classify the FlowPic image. Hu²¹ proposes a network structure called CLD-Net, which combines CNN and LSTM for classifying encrypted internet traffic. Lin²² proposes a method based on flow space-time features of encrypted traffic for effective management of industrial internet of things.

3. METHODS

For any traffic, the spatial and temporal features are consistent in the representation space. This paper proposes a dual encoder architecture that fuses the spatial and temporal features of encrypted traffic using contrastive loss to achieve training objectives based on obtaining spatial and temporal consistent representations in a shared embedding space. The embedding distances of matching spatial and temporal pairs are pulled closer, while the embedding distances of non-matching spatial and temporal pairs are separated, as shown in Figure 1. This method transforms the traffic classification task into a task of fusing spatial and temporal features. By mutually guiding each other, spatial and temporal features maximize the mutual information between different perspectives of the same traffic flow, obtaining a consistent representation as the final traffic representation. Firstly, the input data is copied into two identical copies. In the spatial feature extraction, 1014-dimensional data is processed into a two-dimensional grayscale image to extract spatial features

using a two-dimensional convolutional neural network. In the temporal feature extraction, encrypted traffic data is processed as a single-channel sequence, and one-dimensional convolutional neural network and LSTM are used.

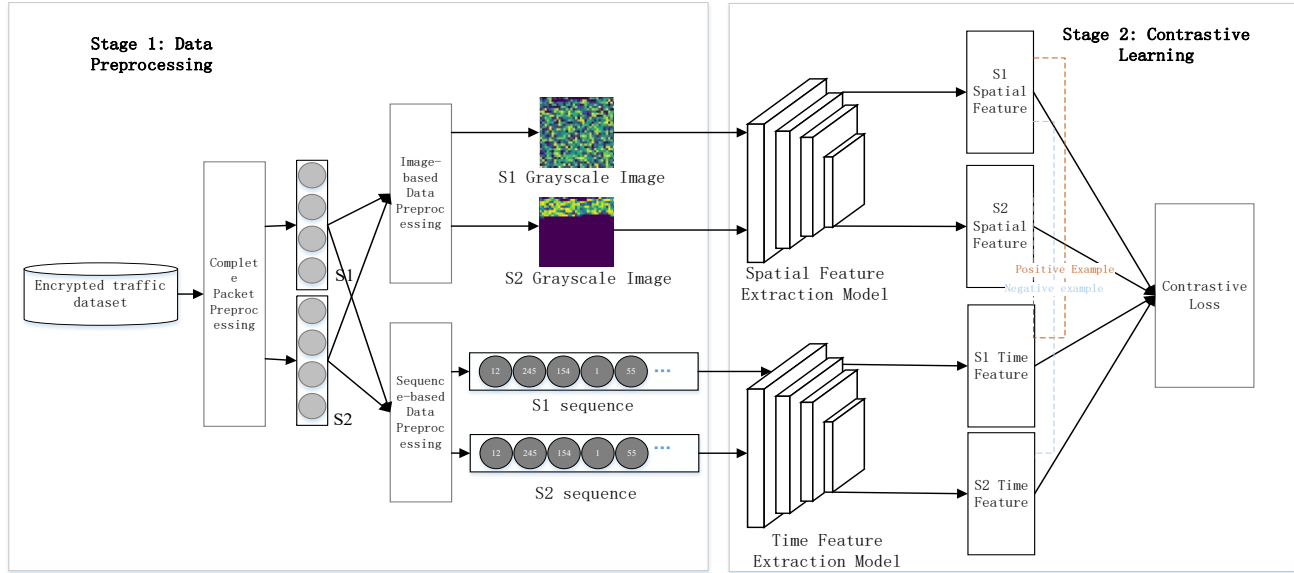


Figure 1. Model architecture.

3.1 Spatial feature extraction

The spatial feature extraction model uses 2D-CNN and consists of four identical blocks, each of which undergoes two convolution operations and two normalization operations. After completion, ReLU is used as the activation function, and the formula is as follows:

$$f_{ReLU}(x) = \max(0, x) \quad (1)$$

After extracting features with four blocks, this paper chooses to use an average pooling layer for pooling operations. The reason is that this method obtains features that are more sensitive to background information and can integrate all information from the feature maps.

$$y_{i,j,k} = \frac{1}{mn} \sum_{p \in P_{i,j}} x_{p,k} \quad (2)$$

where x represents the input feature map, y represents the pooled feature map, m and n represent the size of the pooling window, and $P_{i,j}$ represents the set of positions within the pooling window.

3.2 Temporal feature extraction

As shown in Figure 2, the temporal feature extraction module mainly consists of two parts: a one-dimensional convolutional neural network and LSTM. The one-dimensional convolutional neural network part is composed of three stages, each of which includes one-dimensional convolution, normalization, ReLU activation function, and maximum pooling layer.

This paper first generates more complex patterns in the higher layers by applying one-dimensional convolution operations. During the convolution process, ReLU is selected as the activation function, and normalization is performed after every two convolutions. After one-dimensional convolution, each time step contains more features, which facilitates subsequent feature extraction. After completing the convolution operation, temporal features are extracted from the data flow using LSTM. Moreover, regardless of whether it is the handshake phase or the data transmission phase, the behavior demonstrated in network communication only has forward associations. Therefore, this paper adopts unidirectional LSTM to extract the time features of packets.

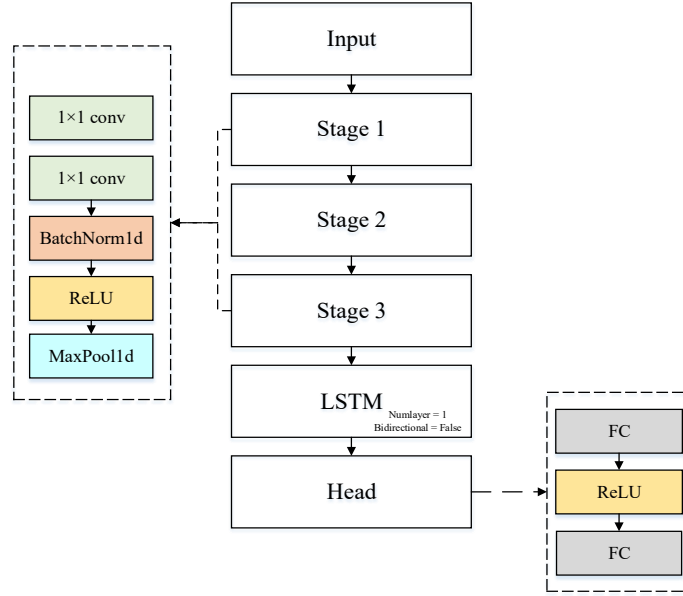


Figure 2. Temporal feature extraction.

3.3 Positive and negative sample selection and loss function

In terms of positive and negative sample selection, since the training goal of this model is to fuse the spatiotemporal features of anchor samples themselves, for anchor samples, their own spatiotemporal features are positive samples. The remaining samples with the same spatial features as anchor samples are negative samples for anchor samples' spatial features. Similarly, the remaining samples with the same temporal features as anchor samples are negative samples for anchor samples' temporal features. The infoNCE Loss is used as the loss function for backpropagation, and the optimal model is obtained by updating gradients and iterating.

$$L(x_i) = -\log \left[\frac{\exp(s_{i,i}/t)}{\sum_{k \neq i} \exp(s_{i,k}/t) + \exp(s_{i,i}/t)} \right] \quad (3)$$

where t is the temperature coefficient. The loss function is calculated as follows:

ALGORITHM 1: Loss function algorithm flow

Input: Sample Vector Matrix V , Hyper-parameter temperature t .

Output: Loss.

Calculating sample similarity **anchor_dot_contrast** = VV^T , divided by the hyper-parameter temperature t .

Subtracting the maximum value from each row, which is the inner product of the sample itself, to improve numerical stability **logits** = **anchor_dot_contrast** - **logits_max**_(dim=1).

Building **mask** Matrix, identifying positive and negative samples, and setting the diagonal to 0, excluding the situation where anchor samples multiply themselves.

Performing computations by converting division to subtraction through logarithmic operations **exp_logits** = **exp(logits)** * **logits_mask**.

Calculating the difference between anchor sample and negative sample **log_prob** = **logits** - **log(exp_logits.sum)**.

Summarized by dividing the sum of products of anchor samples and negative samples by the sum of products of anchor samples and positive samples, and then dividing by the number of positive samples corresponding to the anchor sample.

$$\text{Loss} = (\text{mask} * \text{log_prob}).\text{sum}(1) / \text{mask}.\text{sum}(1)$$

The loss function imposes the same weight penalty τ all negative sample similarities, and through simple calculation of the gradient of the loss function's punishment on different negative sample similarities, it can be seen that the loss function automatically gives more punishment to negative samples that are more similar. The gradient for positive samples is:

$$\frac{\partial L(x_i)}{\partial s_{i,i}} = -\frac{1}{t} \sum_{k \neq i} P_{i,k} \quad (4)$$

The gradient for negative samples is:

$$\frac{\partial L(x_i)}{\partial s_{i,j}} = \frac{1}{t} \frac{\exp(s_{i,j}/t)}{\sum_{k \neq i} \exp(s_{i,k}/t) + \exp(s_{i,i}/t)} \quad (5)$$

4. EXPERIMENTAL RESULT ANALYSIS

4.1 Dataset

In this experiment, we used the ISCX VPN2016 dataset from the Canadian Cybersecurity Laboratory, which is commonly used to study the behavior of VPN traffic and non-VPN traffic. The dataset contains approximately 14,000 samples, including four different types of VPN traffic: OpenVPN, OpenConnect VPN, Cisco AnyConnect VPN, and IPSec VPN, as well as one type of non-VPN traffic. Each category includes multiple application categories such as Facebook, YouTube, Hangouts, Skype, and others. According to different application scenarios, the dataset is divided into three classification tasks: VPN/Non-VPN classification, service classification, and application classification, as shown in Table 1:

Table 1. Classification task.

Classification task	Category
Binary VPN classification	VPN, Non-VPN
Twelve service classification	Chat, Email, Streaming, Voip, VPN: Chat, VPN: File Transfer, VPN: Email, VPN: Voip
Seventeen application classification	AIM chat, Email, Facebook, FTPS, Gmail, Hangouts, ICQ, Netflix, SCP, Sftp, Skype, Spotify, Torrent, Tor, Vimeo, Voipbuster, Youtube

This experiment uses accuracy, precision, recall and F1 score for evaluation and comparison.

4.2 Experimental results and analysis

4.2.1 Ablation experiment. To verify the effectiveness of the encrypted traffic classification model based on spatial-temporal features, an ablation experiment was designed in Twelve Service Classification. As the model in this paper uses a dual encoder structure to extract features separately, we set ablation experiments to extract spatial features only and extract temporal features only. From Table 2, it can be seen that using only spatial features or only temporal features perform worse than the feature fusion model. This result proves that for encrypted traffic, both spatial and temporal features can effectively improve the model, and there is consistency between temporal features and spatial features in a unified encoded space.

Table 2. Ablation experiment results.

Model	PR	RC	F1
Spatial features only	0.822	0.823	0.822
Temporal feature only	0.875	0.875	0.875
Feature fusion	0.982	0.983	0.982

4.2.2 Comparison with existing work. We select the following five existing works for comparison:

Deep packet: Lotfollahi et al.¹⁶ used a convolutional neural network to convolve the sequences and extract local features, then used a recurrent neural network to extract global features.

TSCRNN: Kunda Lin et al.²² proposed a method that combines CNN and BiLSTM to extract spatiotemporal features of encrypted traffic.

1D-CNN: Wang et al.²² processed encrypted packets into 28×28-sized images and compared the performance of 1D-CNN and 2D-CNN on various metrics of encrypted traffic identification.

PERT: He et al.²³ utilized a Transformer model for feature extraction and encoding, and then applied a support vector machine (SVM) for classification.

ET-BERT: Lin²⁴ designed a pre-training task and a reasonable input representation for traffic to demonstrate the effectiveness of pre-trained models.

The results of Binary VPN Classification are shown in Table 3, which indicates that the proposed method achieved higher accuracy when classifying encrypted traffic into VPN and non-VPN categories. It proves that for encrypted traffic, feature fusion can obtain consistent feature representations, and the method maximizes the utilization of unlabeled data, overcoming the disadvantage of previous research models that rely heavily on labeled data.

Table 3. Results of Binary VPN Classification.

Model	Accuracy
TSCRNN	0.977
WANG 1D-CNN	0.999
OURS	0.999

The results of Twelve Service Classification and Seventeen Application Classification are shown in Table 4. The model proposed in this paper outperforms TSCRNN and the deep packet model in terms of AC, PR, RE, and F1 metrics. In Twelve Service Classification and Seventeen Application Classification, it is approximately 5% higher than the aforementioned supervised comparison methods. Taking TSCRNN as an example, although the TSCRNN model also extracts spatial and temporal features, it relies heavily on the labeled data volume of encrypted traffic as it is trained in an end-to-end manner. In contrast, the model proposed in this paper completes pre-training of the model with extensive use of unlabeled traffic and only needs to rewrite the Softmax layer for fine-grained class experiments, using a small amount of labeled data for fine-tuning.

Table 4. Results of Twelve Service Classification and Seventeen Application Classification.

Experimental task	Twelve service classification				Seventeen application classification			
	AC	PR	RC	F1	AC	PR	RC	F1
TSCRNN	0.92	0.93	0.93	0.93	-	-	-	-
Deeppacket	0.9329	0.9377	0.9306	0.9321	0.9758	0.9785	0.9745	0.9765
PERT	0.935	0.940	0.935	0.939	0.823	0.709	0.717	0.700
ET-BERT (flow)	0.9729	0.9756	0.9731	0.9733	0.8519	0.7508	0.7294	0.7306
ET-BERT (packet)	0.989	0.989	0.989	0.989	0.996	0.994	0.994	0.994
OURs	0.982	0.982	0.983	0.982	0.935	0.935	0.935	0.934

In Twelve Service Classification, the proposed model is slightly inferior to ET-BERT (packet), but there is a significant gap between the proposed model and ET-BERT (packet) in terms of application type classification. ET-BERT (packet) adopts the Datagram2Token method to convert encrypted traffic into token units that retain the pattern for pre-training, and proposes two pre-training tasks, Masked BURST Model and Same-origin BURST Prediction, to learn data packet representations from transformational context rather than semantic context.

4.2.3 Experiment with other datasets. To further illustrate the effectiveness of the model proposed in this paper, we conducted validation experiments on two other datasets: USTC-TFC2016 and ISCXTor2016. The ISCXTor2016 dataset contains traffic data from 12,035 Tor relay nodes in 51 different countries, including 8,522 normal nodes and 3,513

malicious nodes. It includes traffic from over 18 representative applications such as Facebook, Skype, Spotify, Gmail, and others, across 8 categories.

Table 5. Experimental results in USTC-TF2016 and ISCXTor2016.

Datasets	USTC-TFC2016				ISCXTor2016			
	AC	PR	RC	F1	AC	PR	RC	F1
TSCRNN	-	0.9870	0.9860	0.9870	-	0.9490	0.9480	0.9480
PERT	0.9909	0.9911	0.9910	0.9911	0.7682	0.4424	0.4446	0.4345
ET-BERT (flow)	0.9929	0.9930	0.9930	0.9930	0.8311	0.5564	0.6448	0.5886
ET-BERT (packet)	0.9915	0.9915	0.9916	0.9916	0.9921	0.9923	0.9921	0.9921
OURS	0.9937	0.9939	0.9939	0.9938	0.9920	0.9920	0.9921	0.9920

As shown in Table 5, in the two datasets, the model proposed in this paper outperforms TSCRNN, PERT, and ET-BERT (flow) in all four metrics, and is on par with ET-BERT (packet).

5. CONCLUSION

This paper proposes a framework for encrypted traffic classification based on the fusion of temporal features and spatial features. By designing different encoders to extract time-series features and spatial features from traffic samples, we obtain the training objective of spatial and temporal consistency representation, and use it to obtain a pre-trained model for encrypted traffic classification. In downstream tasks, only a small amount of labeled data is needed for fine-tuning the Softmax layer to obtain a traffic classification model with good classification performance. Experimental results demonstrate the effectiveness of the proposed model.

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